VANISHING OF EXT^1 AGAINST ALL MODULES VERSUS TWO-TERM PROJECTIVE COMPLEXES, WITH THE NOETHERIAN FINITE VARIANT

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Allowed standard prerequisites (editorial).

- A complex with cohomology concentrated in [-1, 0] has vanishing Ext^1 against every module if and only if H^{-1} = 0 and H^0 is projective. invoked at proof lines 22280-22282. Gives the equivalence of (1) and (2).
- A complex over a Noetherian ring with finite cohomology modules in degrees -1 and 0 is pseudo-coherent. invoked at proof lines 22283-22285. Lets the pseudo-coherent version of the criterion apply.
- For a pseudo-coherent complex the Ext^1 criterion may be tested against finite modules only. invoked at proof lines 22285-22287. Gives the equivalence with (3).

Source reference. Original lemma 0A5P: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17414–17414. Cited at proof lines 22280-22281 for the equivalence of (1) and (2); the conclusion is used inside the counted span.

Source reference. Original lemma 066E: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16517–16517. Cited at proof lines 22283-22284 for the pseudo-coherence of K; the conclusion is used inside the counted span.

Source reference. Original lemma 0G9A: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20136–20136. Cited at proof lines 22285-22286 for the equivalence with (3); the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $K \in D(R)$ with $H^i(K) = 0$ for $i \notin \{-1, 0\}$. The following are equivalent*

(1) *$H^{-1}(K) = 0$ and $H^0(K)$ is a projective module and*
(2) *$\mathrm{Ext}^1_R(K, M) = 0$ for every R-module M.*

If R is Noetherian and $H^i(K)$ is a finite R-module for $i = -1, 0$, then these are also equivalent to

(3) *$\mathrm{Ext}^1_R(K, M) = 0$ for every finite R-module M.*

Proof. The equivalence of (1) and (2) follows from Lemma 0A5P. If R is Noetherian and $H^i(K)$ is a finite R-module for $i = -1, 0$, then K is pseudo-coherent, see Lemma 066E. Thus the equivalence of (1) and (3) follows from Lemma 0G9A. □

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